1 Introduction

We look at the gap-filling problem in metabolic processes.

Here, we have a set of metabolites, and a set of reactions. In steady state, each reaction uses up metabolites at a given rate, and produces metabolites at given rate.

We have a set of experiments where we have measured the metabolites used, and the metabolites produced. We know (or think we know) some of the reactions involved. We wish to work out the other reactions involved. Each potential reaction has a notional 'cost' reflecting the probability it is involved; e.g. this reaction has been seen in similar contexts, or this reaction is from a plant database, so is unlikely to be used.

The decision variables are the *flux*, or throughput of each potential reaction. The constraints reflect the mass balance equations: the amount used and produced must balance with the observed use and output of the experiments.

2 Nomenclature

We have metabolites in the set $M = \{1..m\}$, and reactions in the set $R = \{1..r\}$.

The *flux* v_i determines how active is reaction i. v_i is the principal decision variable of the problem. The 'cost' of reaction i is c_i.

Each reaction involves a number of metabolites. The stoichiometric coefficient a_{ik} determines the amount of metabolite k used or produced in reaction i: a negative coefficient means the metabolite is used, a positive coefficient means the metabolite is produced. There is a linear relationship between the flux and the amounts: for negative stoichiometric coefficients, the total amount of metabolite k used by reaction i is $v_i. - a_{ik}$ (since a_{ik} will be negative). Similarly the amount of metabolite produced is $v_i.a_{ik}$ (since a_{ik} will be positive).

The rate of use of source metabolites – that is, metabolites drawn from the substrate, and not supplied by reactions – is given by s_k. The rate of production of output metabolites is given by p_k.

	R1	R2	R3	R4	R5	Biomass		
Cost	c_1	c_2	c_3	c_4	c_5	-1000		
Flux	v_1	v_2	v_3	v_4	v_5	v_6	LB	UB
Met 1	-1	0	1	1	0	0	-10	0
Met 2	1	-1	0	1	0	-1	0	0
Met 3	0	-1	0	0	-1	0	0	0
Met 4	0	0	-1	-1	0	0	0	0
Met 5	0	1	0	-1	0	1	0	10
Met 6	0	0	0	-1	1	0	0	0

3 Formulation 1

The problem can be expressed simply:

Problem **P1**:

$$\text{minimise} \sum_i c_i v_i \tag{1}$$

subject to

$$\sum_i v_i a_{ik} \quad = \quad p_k - s_k \quad \forall k \in M \tag{2}$$

Note, we add dummy reactions in D to R, each of which produce or consume one metabolite. These ensure that the problem is feasible, allowing an arbitrary amount of any metabolite to either appear or disappear if required. The cost for use of these reactions is, of course, set very high.

Note also that one chosen reaction will be a biomass production reaction. This reaction is kind-of the aim of the process - to produce biomass. The biomass production reaction can have an negative objective coefficient so that it is maximised rather than minimised. The absolute value of the objective coefficient needs to be chosen carefully so that the biomass achieves maximum production given supply restrictions.

4 Dual variables

We can solve this problem using dual variables. We start by solving problem **P1** where only dummy reactions are available. We then examine dual costs of every reaction not currently included. This is a simple calculation involving only the cost of the reaction, and the dual variables of each of the metabolites either used or produced by the reaction (along with the stoichiometric coefficients).

Let λ_k be the dual variable associated with the constraint (2) on metabolite k. The reduced cost of reaction i is

$$\text{RC}_i = c_i - \sum_k \lambda_k a_{ik}$$

We add any reactions with a negative reduced cost to the the LP. We can the solve the new LP, and repeat until no reactions have negative reduced cost.

5 Integer Formulation

The "GlobalFit" system uses an integer formulation, using variables $y_i = 1$ if reaction i is used; 0 otherwise. We use an upper bound on flux (UB_i) for each reaction. The forumlation is then

Problem **P2**:

$$\text{minimise} \sum_i c_i y_i \tag{3}$$

subject to

$$\sum_i v_i a_{ik} = p_k - s_k \quad \forall k \in M \tag{4}$$

$$v_i \leq \mathrm{UB}_i\, y_i \quad \forall i \in R \tag{5}$$

$$y_i \in \{0,1\} \quad \forall i \in R \tag{6}$$

6 Reachability

The formulations in the previous section do not consider the order of production of metabolites.

For instance, consider a reaction r that consumes metabolite A, and produces metabolite B. Another reaction s consumes metabolites B, and produces metabolite A. Neither A nor B are available in the substrate (i.e. $s_A = s_B = 0$). In that case, neither reaction can be used, unless another source of A or B is available. But Formulation 1 will happily use *only* r and s since the use and production of A and B balance out.

This gives rise to the concept of *reachability*, and *depth*.

We say a reaction *enabled* if all the metabolites consumed by a reaction are available. We say a metabolite is available if it is either supplied in the substrate ($s_k > 0$), or a reaction the supplies it is enabled.

We define the 'level' of a reaction as the level it is enabled in the following algorithm. Note: *never* is a constant indicating that the reaction can not be enabled.

The *depth* of the residuals can also be calculated: when all metabolites with p_k ¿ 0 are *available*, we know the residuals can theoretically be produced.

Algorithm *Depth*:

- Make all metabolites k with $s_k > 0$ available
- Set *depth* of all reactions to **never**
- Set *residual_depth* to **never**
- *level* = 0
- *finished* = **false**
- *is_progress* = **true**
- while **not** *finished* **and** *is_progress*
  - *level* = *level* + *1*
  - *finished* = **true**
  - *is_progress* = **false**
  - for each reaction r not yet enabled
    - if all metabolites required by r are available
      - enable reaction r

    - · set *depth* of r to *level*
    - · *is_progress* = **true**
  - ∗ else
    - · *finished* = **false**
- – for each reaction r enabled at this level
  - ∗ for each metabolite i produced by reaction r
    - · make i available
- – if all metabolites k with $p_k > 0$ are available
  - ∗ All residuals can now be produced
  - ∗ set *residual_depth* = *level*
  - ∗ set *finished* = **true**

The variable *is_progress* ensures there is progress at each iteration, or else the algorithm terminates. The variable *finished* is left at **true** when all reactions are enabled.

7 Depth is important

Depth can be used to help ensure that the solution can be implemented in real life.

We note the there must be at least one one reaction from each level in an implementable solution. Once the substrate metabolites are made available, only reactions in level 1 can be triggered. Hence, at least one of those must be used to produce metabolites that are not otherwise available. The same is true at level 2: I can't use any of those reactions until I see the metabolites produced at level 1. But if I want to use reactions in level 3, I must use at least one of the reactions in level 2. This repeats until I reach the depth of the residual metabolites.

This then gives rise to a "valid inequality": at least one equation from each depth must be used.

8 Formulation 2

Let d be the residual-depth as calculated by algorithm *Depth*.

Define $J(r) = j$ for $j \in \{1..d\}$ define the level of reaction $r \in R$. Define LB as a lower bound on a flux for a used reaction. Then we can define Formulation 2 as Formulation 1 plus

$$\sum_{i:J(i)=j} v_i \geq LB \quad \forall j = 1..d \tag{7}$$

9 Dual variables revisited

Let μ_j be the dual variable associated with the constraint (7) on reactions in depth j. The reduced cost of reaction i is now

$$\mathrm{RC}_i = c_i - \sum_k \lambda_k a_{ik} - \mu_j$$

10 Compartments

Cells are modelled to have "compartments" - different areas where reactions take place. Typical compartments are:

- Cytoplasm
- Periplasm
- External
- Outside

External is external to the cell, and the source of normal supply (i.e. the substrate we are growing on). We also model an "outside" compartment, which is also external, but is the site for biomass production and other *residuals* from processes.

We simply expand all reactions and metabolites to be present in/operate in a compartment. Transport reactions move metabolites from one compartment to another (e.g. CO_2 in Cytoplasm can be moved to CO_2 in the Outside.).

11 Multiple experiments

Rather than having a single supply/residual figure, biologists will run several experiments. Having measured outcomes with one substrate (i.e. supply configuration), they may knock out one metabolite from the substrate, and run the experiment again to see if biomass is created.

In this case, the same reaction can be used with different fluxes in each experiment. A different reaction can be used in different experiments, but we want to find the smallest set of reactions that works in all cases. So, if I can explain 2 experiments with a slightly higher flux through 2 reactions, rather than a smaller flux through 3 reactions, I want to choose the solution with 2 reactions.

Expand our nomenclature to include e experiments, and the set $E = \{1..e\}$ of all experiments.

Data is supply/production data (including biomass production) s_k^j and p_k^j on $j \in E$.

We now need flux variables v_i^j, for $j \in E$. We still use an overall flux of v_i, and ensure that v_i takes the maximum value amongst v_i^j values.

Problem **P2**:

$$\text{minimise} \sum_i c_i v_i \tag{8}$$

subject to

$$\sum_i v_i^j a_{ik} = p_k^j - s_k^j \quad \forall k \in M, j \in E \tag{9}$$

$$v_i^j \leq v_i \quad \forall i \in R, j \in E \tag{10}$$